Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: **NerdNinzas**

b. Team leader name: **Prabal Verma**

c. Problem Statement: **Manufacturing: Intelligent Operations & Industrial Efficiency**

OUR SOLUTION

BatchGuard AI

Your factory, watched by an agent that explains every alarm and calls the right person.

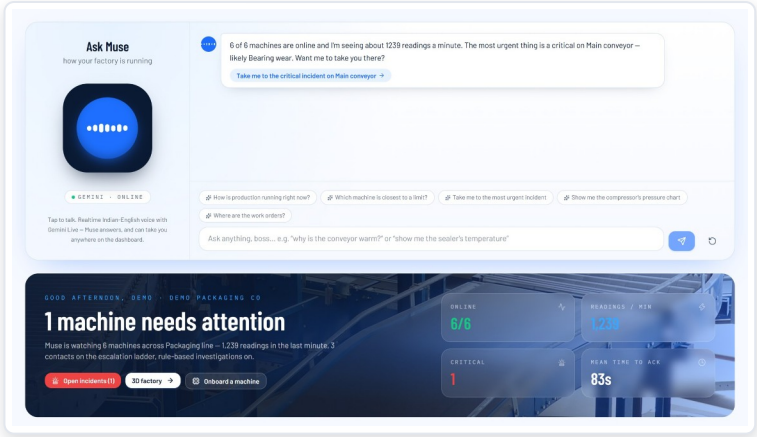


01 / BRIEF ABOUT THE IDEA

An AI plant manager who read every manual and never stops watching

Small factories run on old machines and one stretched maintenance lead. BatchGuard gives them Muse: an agent that turns a machine's PDF manual into live monitoring, works out what is going wrong from the data, and phones the right person before the line stops.

01	**Reads**	Upload the manual. Gemini pulls out every limit with the page it came from, and a person signs it off.
02	**Watches**	Machines post a reading every second. Limits, trends and anomalies are checked on each one, no LLM in the way.
03	**Investigates**	When something drifts, a Google ADK agent gathers evidence, tests three to five explanations and ranks them.
04	**Calls**	If it turns critical, Muse rings the on-call engineer, reads a 20-second brief and takes the acknowledgement.
05	**Learns**	Fixes, photos and verdicts go back into the history the agent reads next time.



WHAT THE ON-CALL ENGINEER HEARS

Muse Main conveyor vibration is climbing toward its limit. Most likely a worn bearing. Check the drive-end bearing first. Can you acknowledge?

Maintenance lead Acknowledged. I'm there in ten minutes.

01 / BRIEF ABOUT THE IDEA: THE PROBLEM

The know-how exists. It is stuck in a PDF and in one person's head

> **A bearing that has been warming for forty minutes gets noticed when the line stops. Then someone goes looking for the manual.**

A TYPICAL BREAKDOWN IN AN SME PLANT

01 No monitoring

SCADA, MES and condition-monitoring suites are priced and staffed for big plants. A shop with eight machines and one maintenance lead cannot buy, integrate or run them.

02 Knowledge locked away

The manual says the bearing must stay under 85 °C and that heat plus vibration means wear. Nobody typed that into a system, and the one person who knows is on another shift.

03 Found late, fixed slowly

Threshold alarms fire after the limit is crossed and never say why. Diagnosis is guesswork, and downtime runs to hours.

WHY NOW

Gemini reads a 60-page scanned manual into structured limits with page citations.

Gemini Live makes a natural phone call with tool use practical, in Indian English.

Google ADK makes a multi-stage evidence pipeline a days-long build, not a year.

Cloud Run + managed data remove the ops work, so a plant costs about $100 a month.

BUILT FOR

- SME plants with 5 to 50 machines
- Maintenance leads who need to know what to check first
- Integrators setting up many plants

02 / OPPORTUNITIES: HOW IS IT DIFFERENT?

Others show you data. We tell you what is wrong and who is fixing it

	INDUSTRIAL IOT PLATFORMS	PLC THRESHOLD ALARMS	"CHAT WITH YOUR PDF"	BATCHGUARD AI
Setting up a machine	🔴 Weeks of integration, priced per data point	🟠 Each limit hand-programmed	🟢 Minutes, but never connected to the machine	🟢 **Minutes: upload the manual, verify, get a key**
Sees the live machine	🟢 Yes	🟢 Yes, on the HMI	🔴 No	🟢 **Every second, per-machine API, live stream**
Warns before the limit	🟠 Through add-on analytics	🔴 Only after it is crossed	🔴 No	🟢 **Trend, anomaly and a time-to-limit forecast**
Explains why	🔴 Charts a person must interpret	🔴 Never	🟠 Knows the manual, not the data	🟢 **Ranked causes, evidence for and against, manual page**
Gets a person to act	🟠 Alerts and email	🔴 A horn or a banner	🔴 No	🟢 **Email, then a phone call, then the next contact**
Fits an SME	🔴 Specialist staff and licences	🟢 Cheap, but blind	🟢 Cheap	🟢 **About $100 a month, nothing to run**

Only BatchGuard closes the loop from a raw reading to a person on the phone who knows what to check.

02 / OPPORTUNITIES: HOW IT SOLVES THE PROBLEM, AND OUR USP

Five things you will not find together anywhere else

HOW IT SOLVES THE PROBLEM

NO MONITORING

Any machine with a sensor or a PLC gateway posts readings to its own keyed endpoint. Cloud Run and managed Google data services mean there is nothing for the plant to operate.

KNOWLEDGE LOCKED AWAY

Gemini turns the manual into verified limits and knowledge passages, fault tables and maintenance intervals, that the agent cites by page.

FOUND LATE, FIXED SLOWLY

Deterministic checks flag drift on every reading. The ADK agent returns ranked causes with evidence and the first check, then phones the on-call contact.

UNIQUE SELLING POINTS

1. **Manual to monitoring in minutes**
   Gemini reads the manual. A person verifies every limit before anything goes live.
2. **Every number is checked**
   A guardrail matches each number in a report against stored readings, limits and manual pages, or marks it unverified.
3. **It picks up the phone**
   Gemini Live calls the on-call engineer. Saying “acknowledged” stops the escalation and the shutdown timer.
4. **You can talk to the factory**
   Ask out loud how the line is running. Muse answers with live data and highlights the chart it means.
5. **It never goes blind**
   Detection needs no LLM, and a rule-based investigator takes over if Gemini is unavailable.

03 / LIST OF FEATURES OFFERED BY THE SOLUTION (1 OF 2)

The monitoring loop: from a PDF manual to a phone call

1

Onboard from the manual

- Gemini 2.5 Flash reads the PDF with a strict JSON schema
- Limits, units, sample rate, page, exact snippet, confidence
- Nothing goes live until a person verifies it
- Device key plus curl, Python and Node samples

2

Per-machine telemetry API

- POST /v1/telemetry with an X-Device-Key
- Parameter and unit checked against the spec
- Duplicates dropped by reading_id
- Live server-sent events to every dashboard

3

Four detection layers

- Limits: warning and critical
- Trend: critical projected within 10 min
- Anomaly: $|z| \geq 3.5$ against its own baseline
- Staleness: a silent machine is an incident

4

Time-to-limit forecast

- Re-fitted every 5 s for each parameter
- Projects time to warning and to critical
- “Critical in 12 min” on cards and charts
- Health score from 0 to 100

5

Incident engine

- One open incident per machine
- Five-minute cooldown per parameter
- Two parameters in warning make it critical
- Every alert attempt is logged

6

Escalation ladder

- Warning: dashboard and email
- Critical: plus a Gemini Live phone call
- No answer in 2 min: the next contact
- No ack in 5 min: Muse stops the machine

6.4 ms median time to ingest and check a reading

22 s from a bearing starting to fail to a warning

0 LLM calls anywhere in the detection path

03 / LIST OF FEATURES OFFERED BY THE SOLUTION (2 OF 2)

What makes Muse more than an alarm

07 Investigation agent

A six-stage Google ADK pipeline with seven read-only tools. The report has a summary, timeline, ranked causes, ordered checks with safety notes, missing data and a 20-second call script.

08 Number guardrail

Any number the agent cannot trace to stored readings, limits, neighbours or manual pages becomes [unverified] and is listed.

09 Fallback investigator

If Gemini fails twice, a rule-based investigator with a fault-signature library writes the same report shape.

10 Phone escalation

Five call flows run on Dograh over a TeleCMI line. Transcript, recording and the extracted outcome update the incident.

11 Muse copilot

Streaming chat and realtime Gemini Live voice. Ask for anything and Muse navigates there and spotlights it.

12 Photo evidence

A technician's photo is read by Gemini vision against the ranked causes and can re-run the investigation.

13 Work orders

A Kanban per machine. One click turns Muse's checks into orders, prioritised by severity.

14 Live 3D factory and line builder

A 3D hall that mirrors machine status, and a node editor whose upstream and downstream links drive cross-machine reasoning.

15 Observability and evaluation

Latency by route, ingest rate, every agent tool call, and a blind test that scores detection delay and cause accuracy.

04 / PROCESS FLOW DIAGRAM (1 OF 3)

One reading's journey, from a sensor to someone's phone

DETERMINISTIC, EVERY READING, MILLISECONDS

01	02	03	04
Onboard	**Ingest**	**Detect**	**Open incident**
Upload manual	POST /v1/telemetry	Limits	One per machine
Gemini extracts limits	Checked against spec	Trend to time-to-limit	Five-minute cooldown
Person verifies	Stored, streamed live	Anomaly (z-score)	Severity ladder
Device key issued	Pub/Sub to BigQuery	Staleness, forecast	Health score

AI, PER INCIDENT

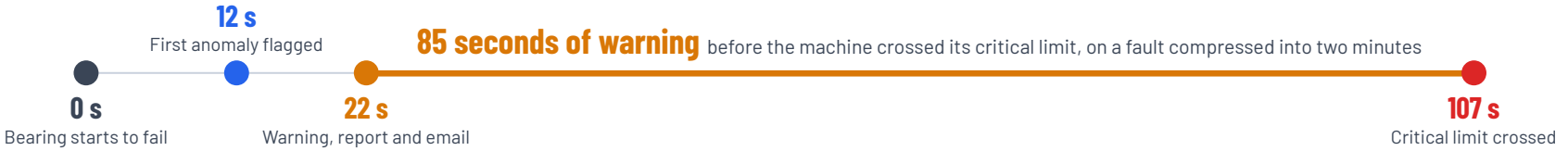
05
Investigate
Evidence in parallel
Hypotheses, evaluation
Checks, report
Number guardrail

PEOPLE

06	07
Escalate	**Act and learn**
Dashboard and email	Ack spoken on call
Gemini Live call	Work orders
Next contact at 2 min	Photo evidence
Stop at 5 min if no ack	Resolve with verdict

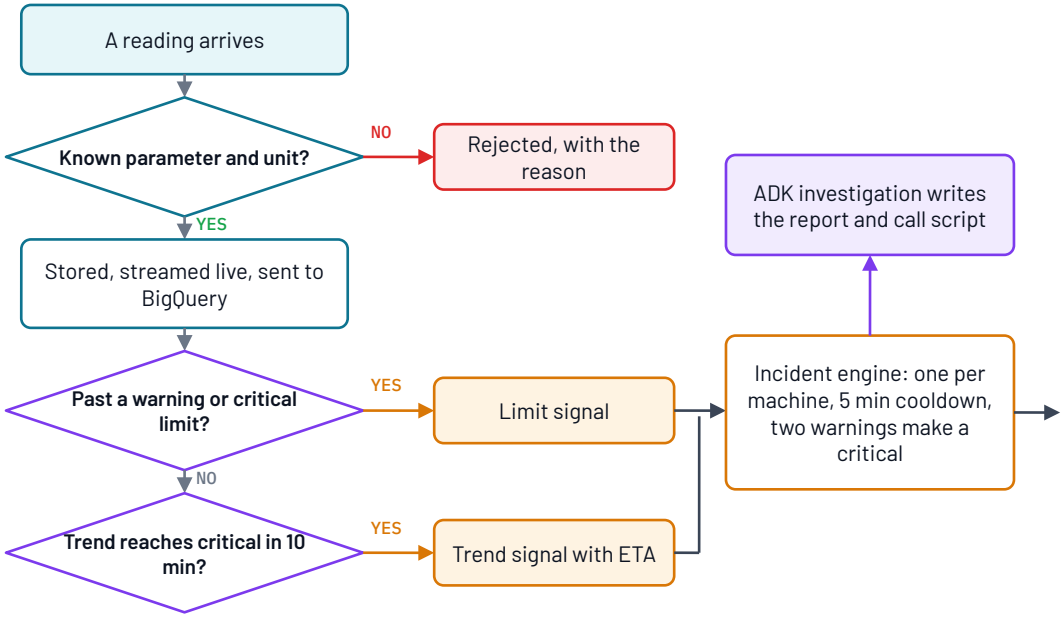
Photos and verdicts re-run the investigation

MEASURED ON THE PROTOTYPE: BEARING WEAR ON A CONVEYOR (MEDIAN OF RUNS)



04 / PROCESS FLOW DIAGRAM (2 OF 3)

Every reading gets the same checks. Severity decides who hears about it



Also on every reading: a z-score of 3.5 or more against the machine's own baseline logs an info event, and the forecast refreshes every 5 s.

SEVERITY LADDER

Info — One anomaly, inside limits

A dashboard event. No incident, no noise.

Warning — A limit or a trend

Dashboard and email, plus the report when the agent finishes.

Critical — Critical limit, or two warnings at once

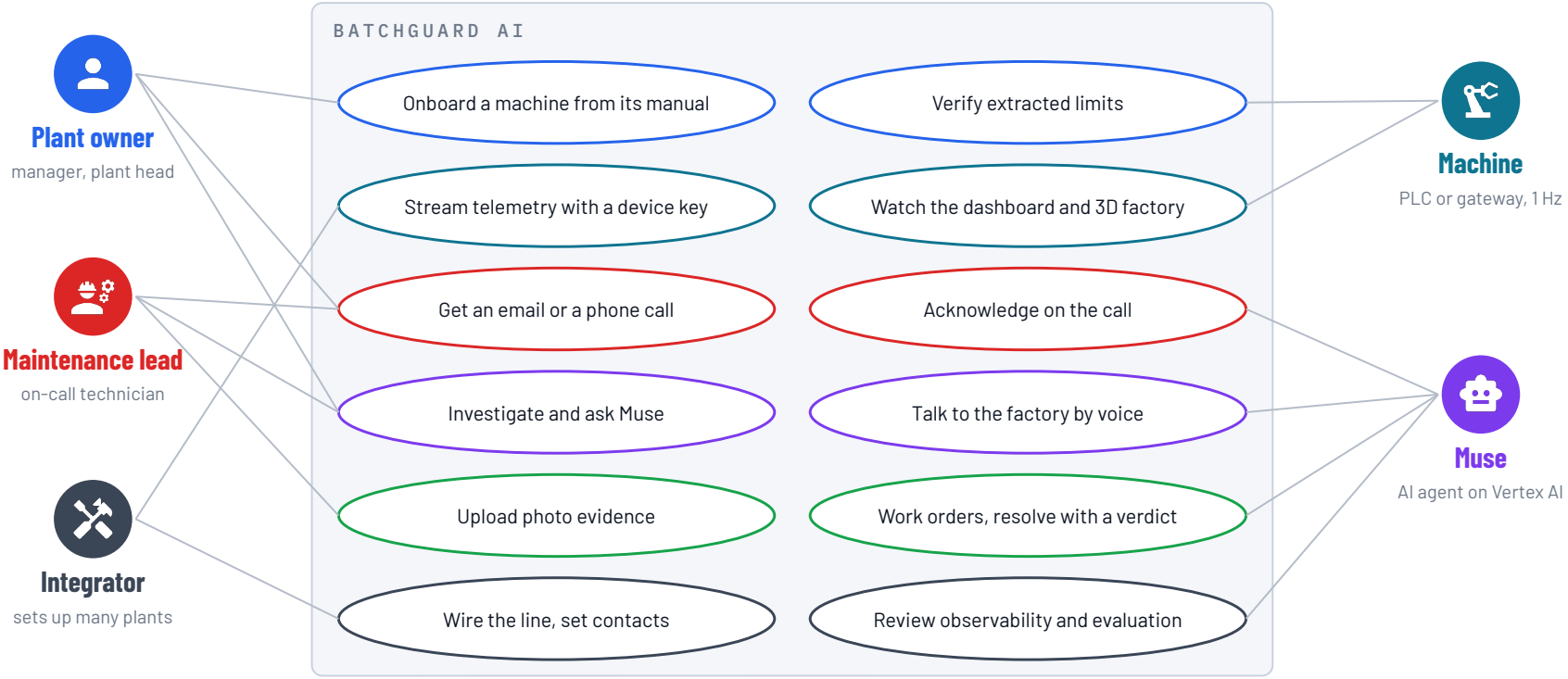
Everything above, plus a Gemini Live call to contact #1, and the shutdown timer starts.

Emergency — No acknowledgement in 2 min, or an E-stop

Call the next contact every 2 min. No one in 5 min: Muse shuts the machine down.

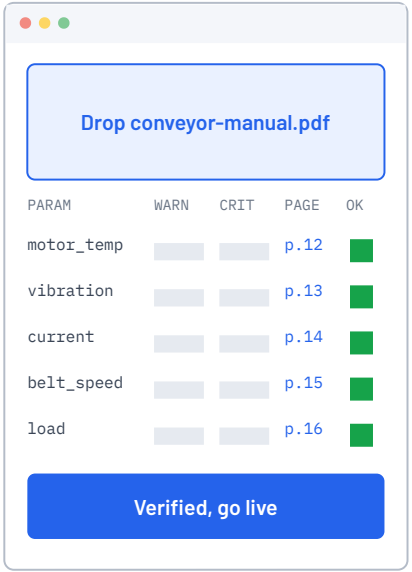
04 / PROCESS FLOW DIAGRAM (3 OF 3): USE CASES

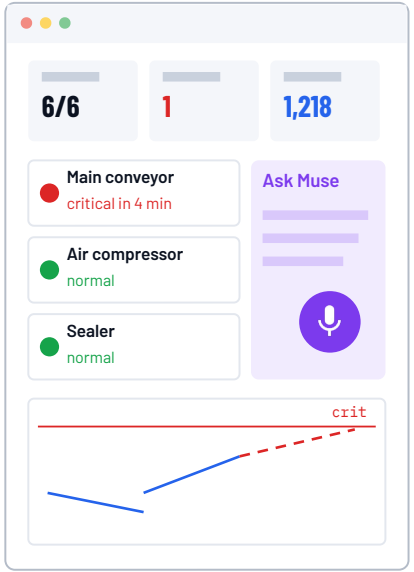
Who uses BatchGuard, and for what

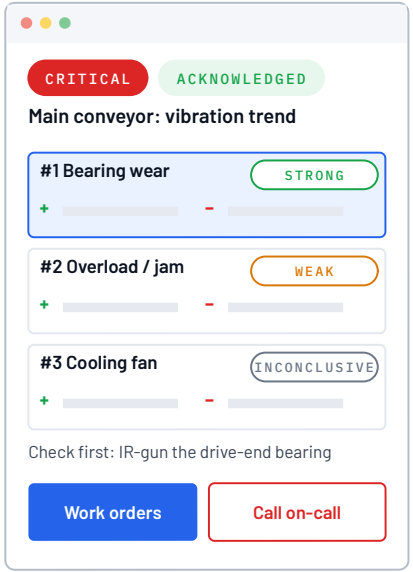


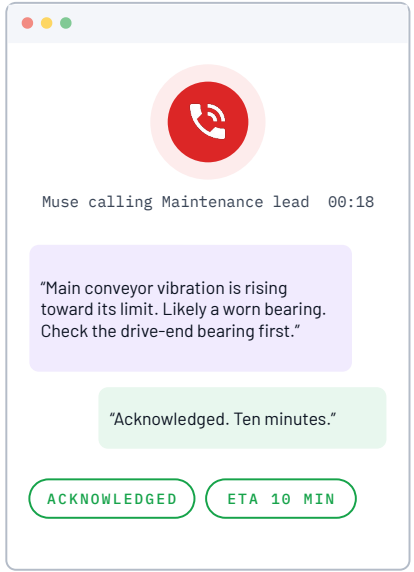
05 / WIREFRAMES AND MOCK DIAGRAMS

Four screens carry the whole journey









01 Onboard

Upload the manual, verify every limit

02 Overview

The live line, forecasts and Ask Muse

03 Incident

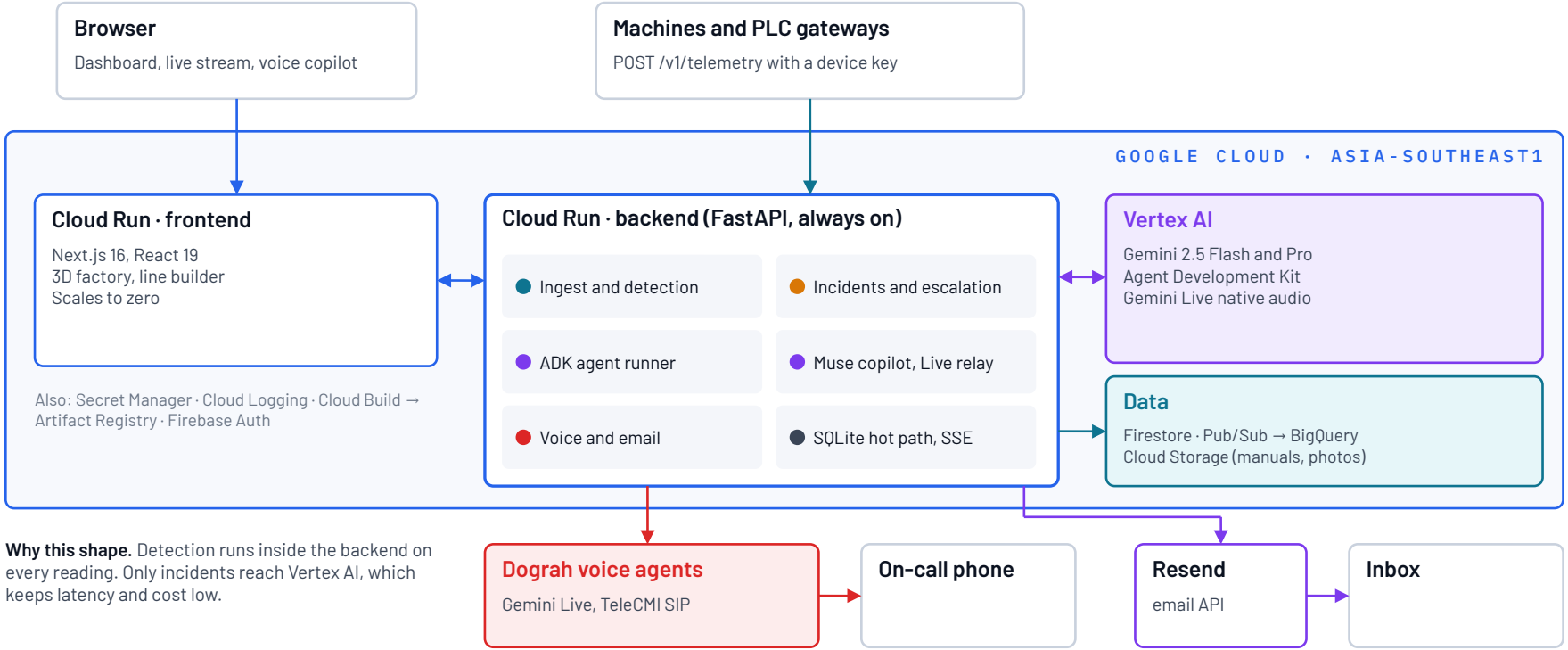
Ranked causes, evidence, checks

04 The call

Muse phones the on-call engineer

06 / ARCHITECTURE DIAGRAM OF THE PROPOSED SOLUTION (1 OF 2)

Two Cloud Run services. Vertex AI does the thinking



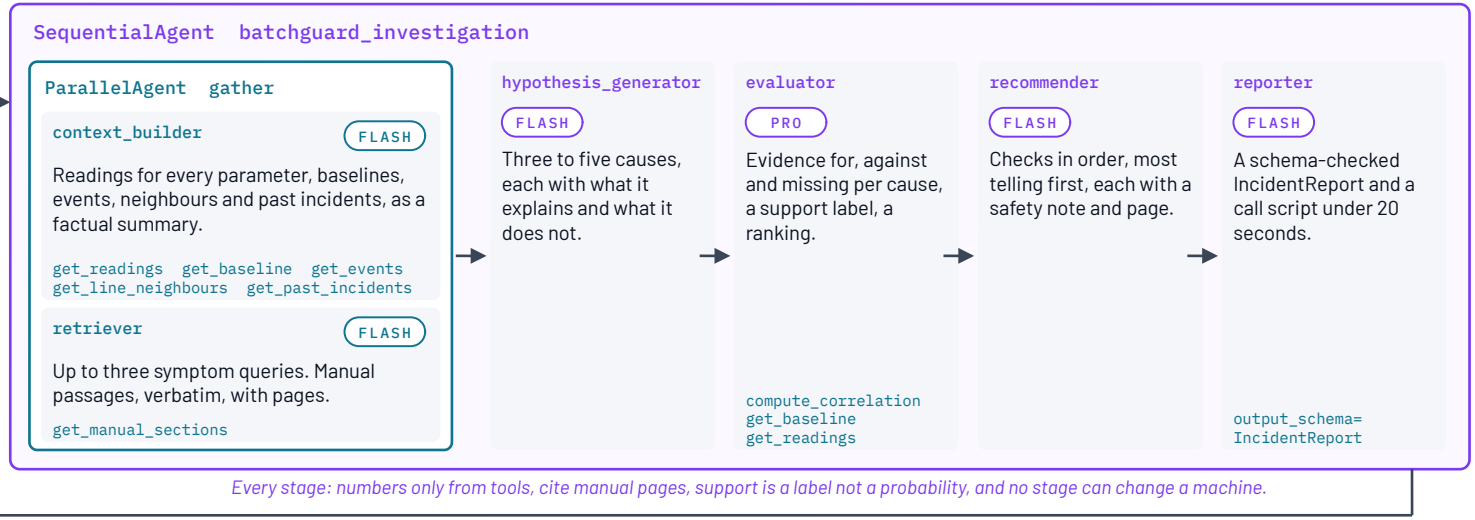
Why this shape. Detection runs inside the backend on every reading. Only incidents reach Vertex AI, which keeps latency and cost low.

06 / ARCHITECTURE DIAGRAM OF THE PROPOSED SOLUTION (2 OF 2)

The investigation is a pipeline, not a prompt

TRIGGER

An incident opens or escalates. One run per incident at a time.



Every stage: numbers only from tools, cite manual pages, support is a label not a probability, and no stage can change a machine.

Number guardrail

Each number is matched against recent readings, limits, manual pages and neighbour values within 0.5 %. Anything else becomes [unverified].

Store and broadcast

The report is saved, pushed live to every dashboard and mirrored to Firestore. The tool trace is kept per run.

Notify

A report-ready email goes out. The call script and first check fill the phone call.

Fallback

If Gemini fails twice, a rule-based investigator uses the same tools and a fault-signature library.

07 / TECHNOLOGIES TO BE USED IN THE SOLUTION

Built on the Google Cloud stack the brief asked for

Gemini on Vertex AI	●	2.5 Flash reads manuals, gathers evidence, chats, reads photos and call outcomes. 2.5 Pro weighs the hypotheses.
Vertex AI	●	Agent Development Kit runs the six-stage investigation. Gemini Live native audio powers the voice copilot.
BigQuery	●	Stores every reading through a Pub/Sub BigQuery subscription, for long-window history and analysis.
Cloud Storage	●	Keeps uploaded manuals and datasheets, and technicians' photo evidence.
Vision and image analysis	●	Gemini multimodal reads scanned manuals and photos against the ranked causes.
Pub/Sub	●	The telemetry topic carries every reading from ingest to the warehouse.
Dataflow	●	Next: windowed feature pipelines when a customer runs many sites.
Cloud Run	●	Runs the frontend (scales to zero) and the backend (always on), built by Cloud Build.

● in the prototype ● planned

ALSO ON GOOGLE CLOUD

Firestore
Secret Manager
Cloud Logging
Cloud Build and Artifact Registry
Firebase Authentication

APPLICATION STACK

Frontend

Next.js 16, React 19, TypeScript, Tailwind v4, three.js and React Three Fiber, React Flow, Recharts

Backend

Python 3.12, FastAPI, Pydantic, Google ADK, google-genai, NumPy

Voice and email

Dograh, TeleCMI SIP, Resend

08 / ESTIMATED IMPLEMENTATION COST

About $100 a month runs a plant. One hour of downtime costs more

≈ $100

per plant, per month, on Google Cloud list prices

Six machines, 21 parameters at 1 Hz (about 54 million readings), around 60 investigations, 40 call minutes and two hours of voice copilot.

≈ $15 to 25 for each additional plant. One backend serves many plants.

$0 on servers, SCADA or MES licences. Existing PLCs post over HTTPS.

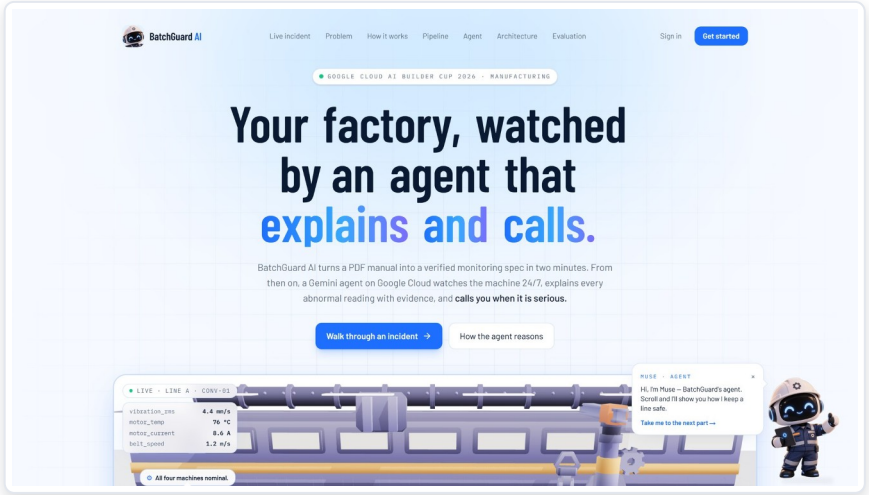
MONTHLY BREAKDOWN, USD

Cloud Run backend	1 vCPU, 1 GiB, always on, Singapore	50 – 70
Cloud Run frontend	Scales to zero, mostly free tier	0 – 3
Vertex AI investigations	~60 runs, ~45k Flash + ~10k Pro tokens each	3 – 5
Vertex AI other calls	Manual extraction, chat, vision, call outcomes	2 – 5
Gemini Live audio	Voice copilot and phone agent, ~160 min	3 – 8
Pub/Sub and BigQuery	54M messages, ~11 GB stored	3 – 6
Firestore, Storage, Secrets, Logging	Small volumes	1 – 3
Phone line and voice server	TeleCMI minutes, small VM for Dograh	15 – 30
Email	Resend free tier, 3,000 a month	0
Total, first plant		80 – 130

Prices as of October 2026, before tax. Vertex AI Gemini 2.5 Flash $0.30 / $2.50 and 2.5 Pro $1.25 / $10 per million input / output tokens; Cloud Run instance billing, Tier 2 region. Cost levers: batch tier, context caching, committed-use discounts. Sensors cost extra only where a machine has no PLC or gateway.

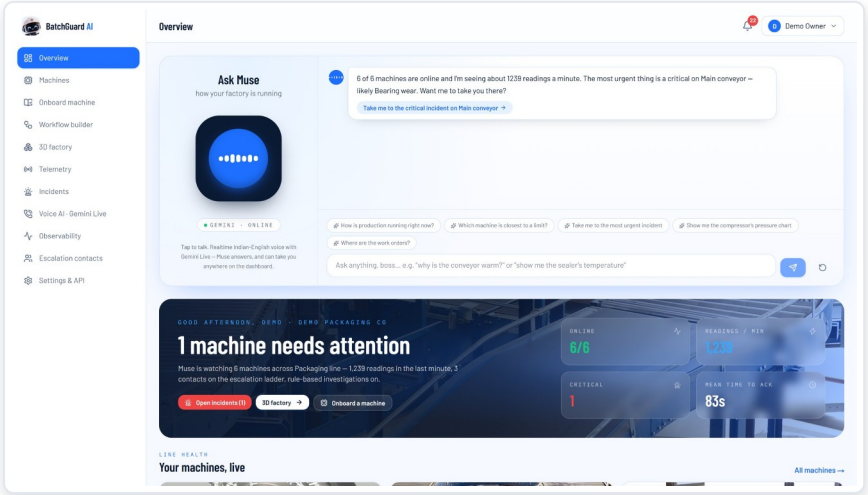
09 / SNAPSHOTS OF THE PROTOTYPE (1 OF 4)

The live prototype, running on Cloud Run



01 Landing page

The story in one scroll: a live incident, the problem, how the agent reasons, the architecture and our evaluation, with Muse as the guide.

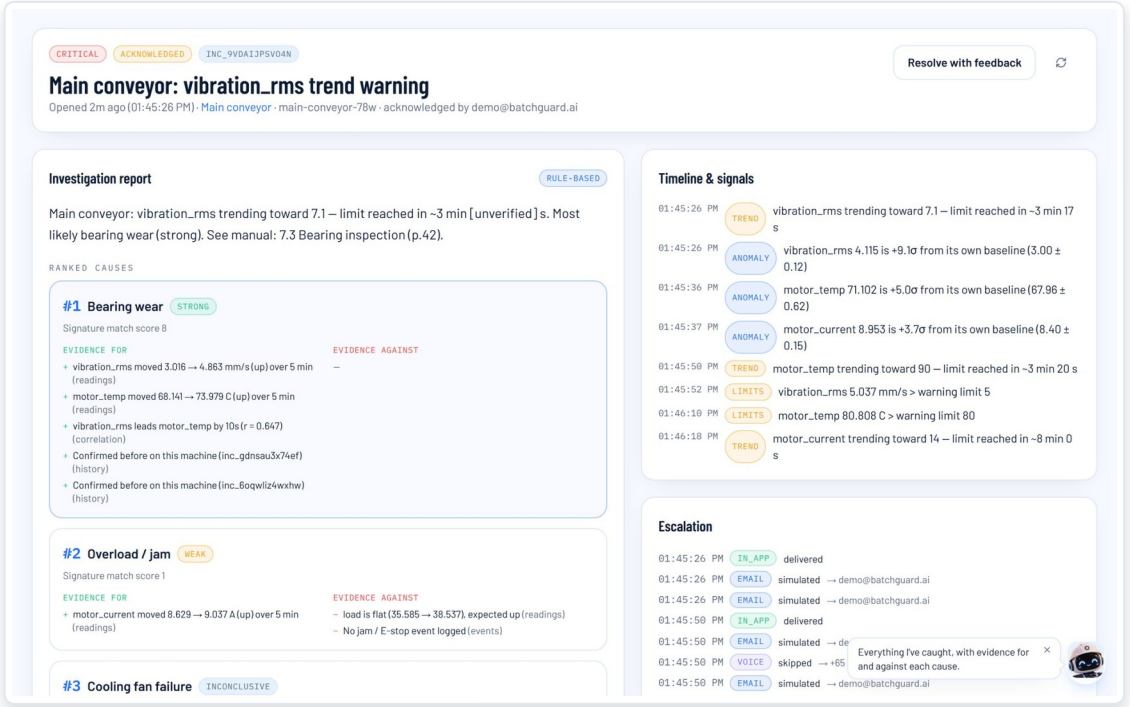


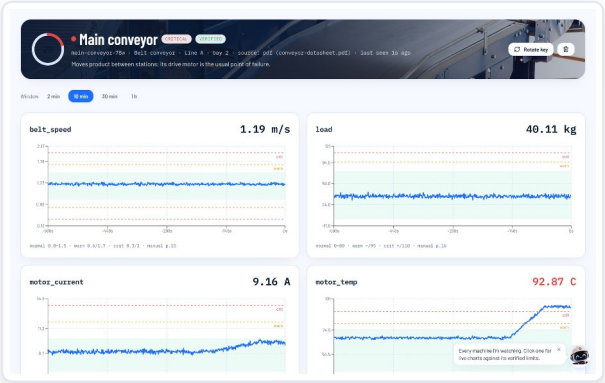
02 Overview with Ask Muse

Muse opens with a summary built from live data. Below it: machine health, readings per minute, forecasts and open incidents.

09 / SNAPSHOTS OF THE PROTOTYPE (2 OF 4)

An incident, investigated





03 Incident report

Ranked causes with evidence for and against, the manual page, the first checks, and every signal on a timeline.

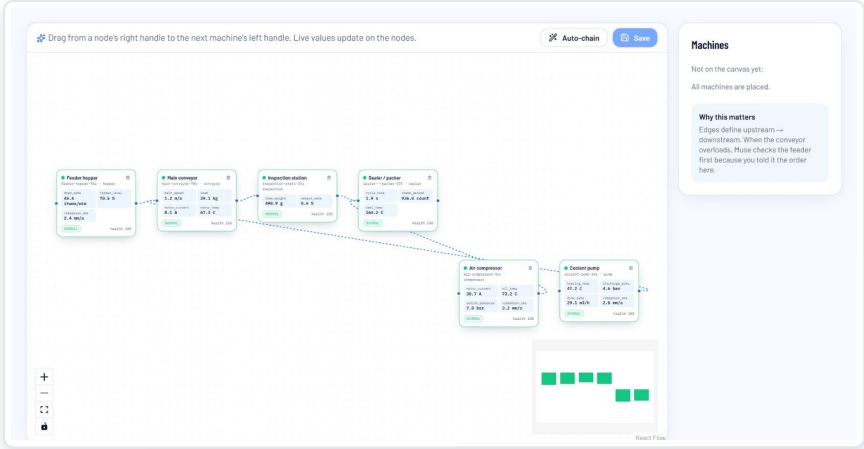
04 Machine page

Every parameter against its verified bands, with the manual page and a time-to-limit forecast.

09 / SNAPSHOTS OF THE PROTOTYPE (3 OF 4)

The factory, live





05 3D factory

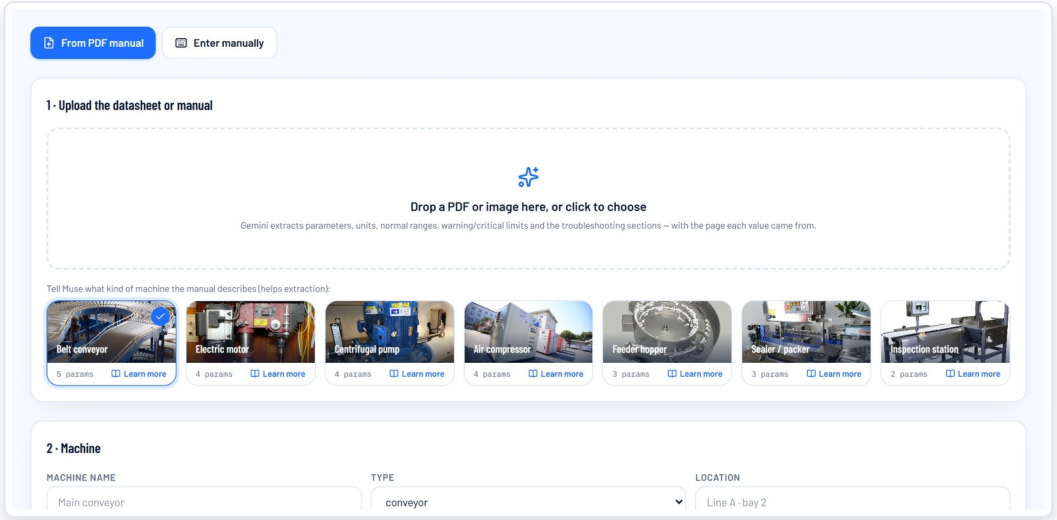
A React Three Fiber hall with live readings over each machine. Colours follow normal, warning and critical.

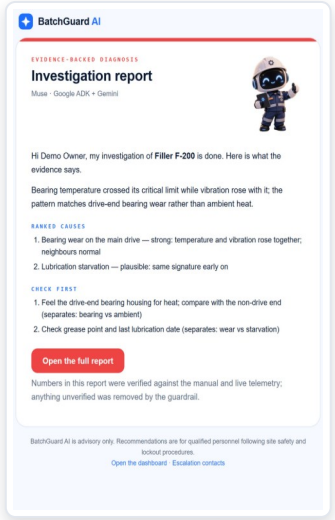
06 Workflow builder

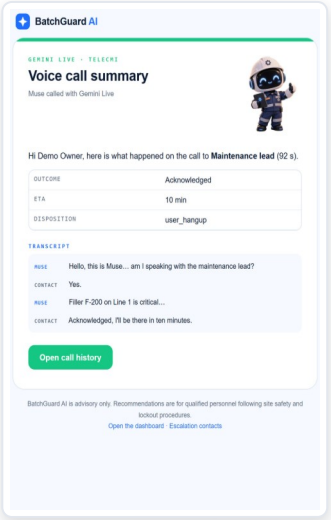
Drag machines into a line. The links set what the agent checks across machines, like a feeder over-feeding the conveyor.

09 / SNAPSHOTS OF THE PROTOTYPE (4 OF 4)

Onboarding and the emails Muse sends







07 Onboard a machine

Upload the manual or pick a machine type. Gemini fills in the parameters, limits, page and snippet. Verify each row, then copy the device key and code samples.

08 Report and call emails

The likely cause, the first check, and what was said on the call.

10 / PROTOTYPE PERFORMANCE REPORT / BENCHMARKING (1 OF 2)

16 injected faults. All 16 caught. The right cause first in 13

16/16
faults detected

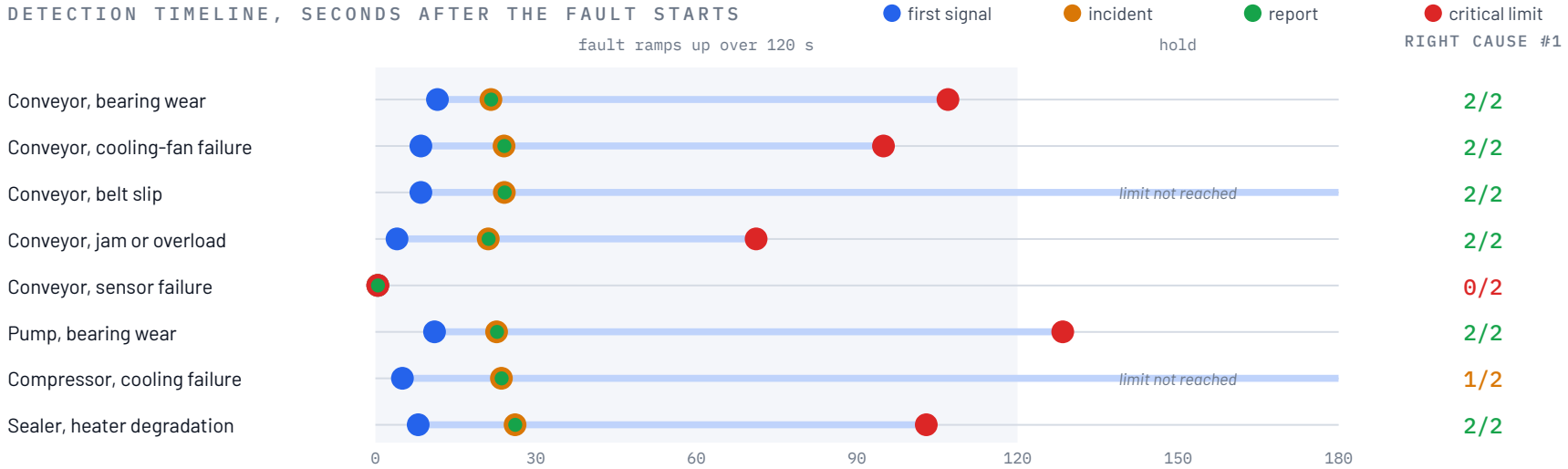
13/16
right cause ranked first (14/16 in the top three)

8 s
median time from fault to first signal

77 s
median warning before the critical limit

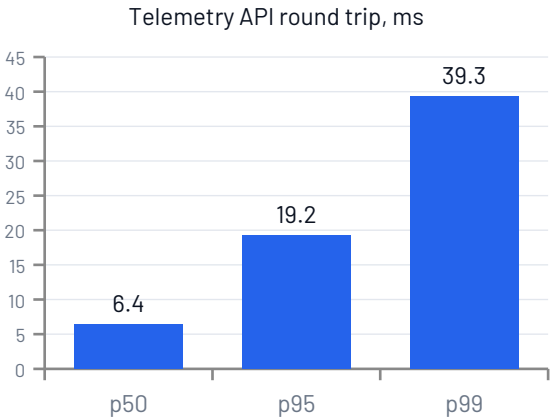
0
false incidents in 53 machine-minutes of normal running

DETECTION TIMELINE, SECONDS AFTER THE FAULT STARTS



10 / PROTOTYPE PERFORMANCE REPORT / BENCHMARKING (2 OF 2)

Fast enough to run a whole plant on one small container



16 machines, 5 parameters each, at 1 Hz: 80 readings a second, each validated, stored and run through detection.

6.4 ms

median ingest round trip at a plant's normal load

5.7×

headroom: one container sustains ~120 readings/s, the demo plant needs 21

0.03 s

median from incident to finished report, fallback investigator, local run

14/16

runs where the trend layer raised the first warning, before any limit was crossed

About the Gemini path. On Cloud Run the six-stage ADK pipeline runs on Vertex AI, and each run's duration and tool calls appear on the Observability page. This benchmark ran on a laptop without Vertex credentials, so it measures detection and the fallback investigator.

WHAT THE TESTS TAUGHT US

- Misses: sensor failure twice (a reading that drops to zero was read as belt slip) and one compressor run that blamed a belt it does not have. Fix: a step-change signature, and skip causes whose key parameter a machine lacks.
- The guardrail marked 20 derived numbers (mostly ETA seconds) as [unverified] in 11 of 16 reports. It is strict on purpose; values our own tools compute will be whitelisted.
- 242 of 254 info events in normal running came from a counter (items_packed). They never open incidents, and counters will be excluded from the z-score check.

11 / ADDITIONAL DETAILS AND FUTURE DEVELOPMENT

Next: more machines, more languages, earlier warnings

Next 3 months

- OPC UA, Modbus TCP and MQTT gateway connectors
- WhatsApp and SMS escalation; Hindi, Tamil, Bahasa and Japanese voice
- The agent reads resolved incidents with confirmed causes as evidence
- Postgres or AlloyDB for many tenants

3 to 9 months

- Dataflow streaming features (rolling RMS, spectra) into BigQuery
- Vertex AI remaining-useful-life models trained on BigQuery history
- Camera-based visual quality inspection linked to reject rate
- Work orders synced with SAP PM and IBM Maximo

9 to 18 months

- Energy per unit, compressed-air leaks and idle power alerts
- Benchmarks across sites for integrators and plant groups
- Spare-parts demand forecast from failure history
- Lock-out and tag-out checklists inside the call

Advisory by design

Its one action is a protective stop when nobody acknowledges. An operator restarts.

Keys and secrets

Device keys are hashed and rotatable. Partner keys live in Secret Manager.

A person signs off

Limits go live only after verification. Auto-dial is off by default.

Built in JAPAC

Indian-English voice today. Call flows and emails localise per plant.

12 / PROVIDE LINKS TO YOUR

See it running

GITHUB PUBLIC REPOSITORY

github.com/NerdNinzas/batchguard.ai

Next.js frontend, FastAPI and ADK backend, call flows, deploy script, IDEA.md and SOLUTION.md.

DEMO VIDEO

youtu.be/IxMaqhoitiU

Setup from a manual, a tour, then three live cases: all fine, a breakdown acknowledged by phone, and no answer until Muse shuts the line down.

FINAL PRODUCT

batchguardai.live

Sign in with demo@batchguard.ai / demo1234. Demo Packaging Co, six machines, live telemetry.

VIRTUAL FACTORY SIMULATOR

batchguard-sim.web.app

Streams real telemetry into the demo account. Turn the dials or inject a fault, and watch Muse react.

Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.

Thank you

